

Nothing to Route On: A Measured Information Ceiling Explains Routing Absorption in Hybrid Mamba-Attention Language Models

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(local preview rendered from HSPMN_v5_draft.tex; arXiv compiles the LaTeX source)

Abstract

Small hybrid language models pair an attention stream with a state-space stream and add a learned gate that decides, token by token, how much each stream contributes. We ask a simple question: does the gate actually do anything? Across 134 training runs at 140M and 350M parameters on a single consumer GPU, the answer is no, and we can measure why.

First, the architecture. A block-level dual-stream design with a learned router loses to plain head-level fusion by +19.5% perplexity at matched parameters and ingredients, and we trace the loss to the router itself: 6 of 12 layers have a gate that is zero for every token and two more route under 0.03% of tokens. Second, the gate. The learned gate fails to beat a frozen random gate at 140M and again at 350M (3 seeds each), reproducing the routing-absorption result of Aquino-Michaels at 11x their scale.

Third, the explanation. We treat the gate as a noisy channel between the token and the event 'the attention stream lowers the loss here', and prove via Fano and Pinsker inequalities that a gate's benefit over an always-on baseline is capped by $\sqrt{2(I + \log_2 - H(s))}$. Realising the full routing benefit needs about 0.72 bits. Trained gates carry 0.004 to 0.006 bits. An optimal probe trained directly on the oracle label reaches only 0.034 bits: 21x short. The routing target is nearly information-free, so no token-local gate of this class can work, trained or not. Ablating the attention stream hurts on 256 of 256 held-out sequences and 98.9% of the benefit variance lies within sequences: the optimal routing policy is simply 'always on', which is the gate-less baseline. The gates fail not because routing is hard to learn but because, on this data, there is nothing to route on. The theory identifies the two levers that could change this, data heterogeneity and stream non-redundancy, and we close by pre-registering both as a falsifiable programme with predictions committed before the results.

Contributions

1. Block fusion retired, with a mechanism. +19.5% PPL loss vs head fusion at matched ingredients; router modally collapsed (6 dead + 2 effectively dead of 12 layers; one saturated layer at gate mean 47, std 316); 19% more forward FLOPs per token.

2. The gate null, replicated at two scales. Learned vs random gate: 143.85 ± 9.27 vs 146.24 ± 7.56 at 140M; 82.35 ± 4.78 vs 83.79 ± 1.37 at 350M (Welch $t = -0.41$). Gate-less beats both. Head fusion's own -3.4% edge over dense at 140M decays to a tie at 350M.

3. A gate-channel bound that predicts the null. Advantage cap $\sqrt{2(I + \log_2 - H(s))}$; full benefit needs ~ 0.72 bits; measured gates carry 0.0059 ± 0.0004 (learned) vs 0.0029 ± 0.0006 (random) bits, capping every gate at $\leq 15\%$ of the ideal benefit.

4. The ceiling: the null is structural. Optimal probes trained on the oracle label reach 0.034 bits (21x below threshold; any token-local gate capped at 24% of ideal). Attention helps on 256/256 sequences; 98.9% of benefit variance is within-sequence. Optimal policy: always on.

5. A pre-registered programme, executed. Routability phase diagram over mixture corpora with recall fraction π in $\{0, .1, .25, .5\}$: the oracle-label ceiling rises monotonically 0.018 $\rightarrow$ 0.139 bits (P1 confirmed) yet stays 5x below threshold, and every gate-vs-random null lands where the bound requires (P2, Welch $p \geq 0.52$ at all π). Stream decorrelation fails to raise the ceiling (P4 falsified). Post-hoc: the parameter-free PC gate beats the learned gate pooled across the diagram (-2.35 PPL, $p = 0.041$) and ties gate-less.

Table 1: 140M results (3 seeds, best LR per variant)

Variant	PPL (3 seeds)	vs dense	tok/s
dense (muP-tuned)	144.01 ± 5.64		83,154
hymba (head fusion, dense attn)	143.42 ± 9.98	-0.4%	66,071
hymba-with-nsa (gate-less)	139.14 ± 8.43	-3.4%	50,582
+ PC gate (parameter-free)	140.09 ± 7.21	-2.7%	49,204
+ param-free select	141.01 ± 9.75	-2.1%	45,764
+ learned gate	143.85 ± 9.27	-0.1%	49,720
+ random gate (A-M control)	146.24 ± 7.56	+1.6%	50,029
ASA alternation	149.29 ± 9.42	+3.7%	61,795
v4-gdn-nsa (block fusion + router)	166.23 ± 15.12	+15.4%	35,578
v4-rwkv7-nsa (block fusion)	169.55 ± 14.03	+17.7%	32,551

Table 2: 350M results (3 seeds each, 983M tokens)

Variant (350M)	PPL by seed 42/1337/2026	mean ± std
hymba-with-nsa (348.8M)	87.38 / 86.99 / 81.60	85.32 ± 2.64
dense, muP (368.8M)	84.23 / 80.18 / 86.82	83.75 ± 2.73
hymba-with-nsa-gated	78.25 / 89.06 / 79.75	82.35 ± 4.78
hymba-with-nsa-randgate	82.93 / 85.73 / 82.73	83.79 ± 1.37

Neither contrast significant (Welch $t = 0.72$ and -0.41). The seed-42 gated run (78.25) looked like a 10% win until seeds 1337/2026 arrived; seed luck, told in full in the paper.

Proposition (gate-channel advantage bound)

With entropies in nats, any routing decision derived from the gate has advantage over chance capped by $(1 - 2 P_{\text{err}}) + \leq \sqrt{2(I(g;s) + \log 2 - H(s))} =: A(I)$, and its benefit over the always-on baseline is capped by $(\sigma_C^2 / 2L) * A(I)$. Realising the full ideal benefit requires $I \geq 1/2 + H(s) - \log 2$ nats, about 0.72 bits for a near-balanced label. Proof: Fano's inequality plus Pinsker's inequality (the chord bound $H_b(q) \leq 2q \log 2$ is false by concavity; Pinsker gives the valid direction). Corollary: at $I = 0$ and balanced s , the cap is exactly zero, recovering the Aquino-Michaels null.

Measured: learned 0.0059, random 0.0029, PC 0.0072 bits at 140M (non-circular control confirms); 0.0040 / 0.0021 at 350M. Advantage caps: 14% / 13% / 15% of ideal. Ceiling probes (linear on h_t : 0.0315; MLP on both streams' outputs: 0.0341 bits) cap ANY token-local gate at 24% of ideal.

Figures

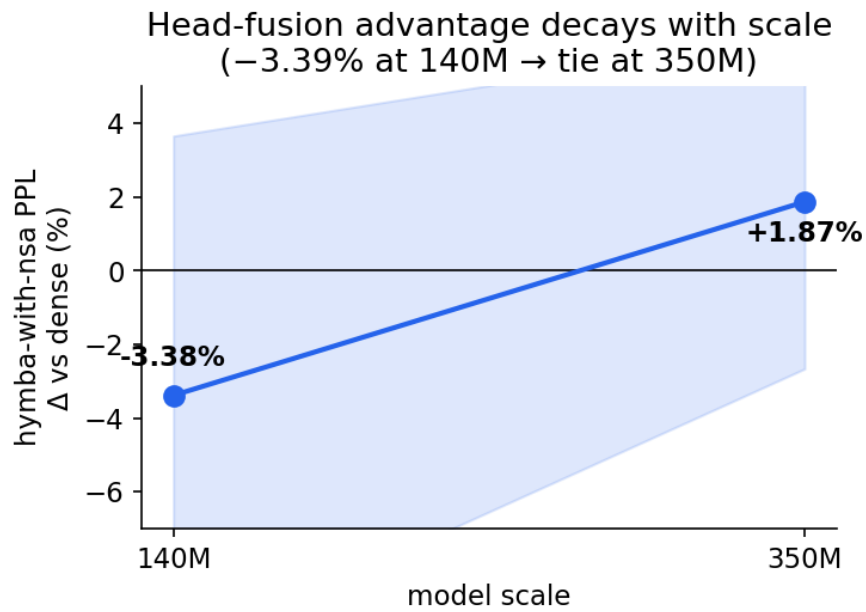


Fig 1: Head fusion's PPL advantage decays from -3.4% (140M) to a tie at 350M.

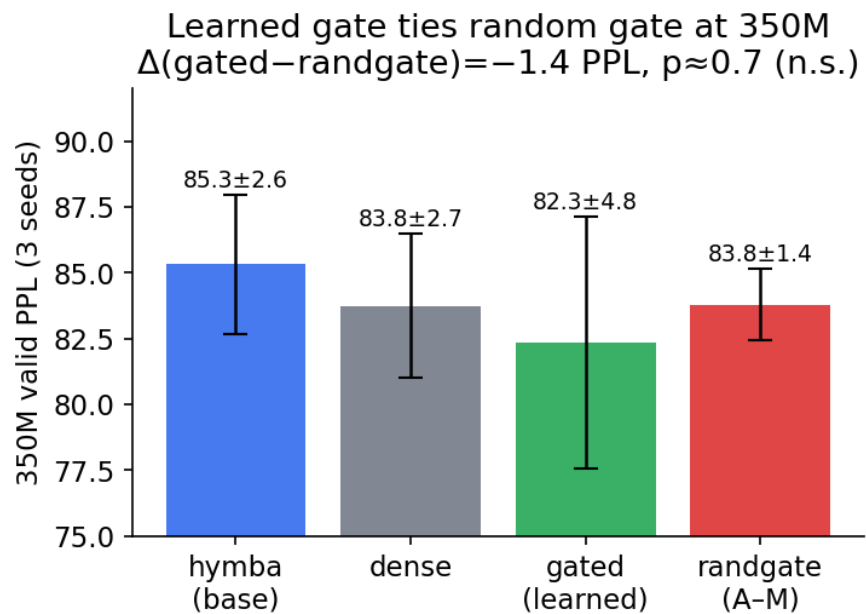


Fig 2: Learned vs random gate at 350M: means tie; learned is the higher-variance way to get nothing.

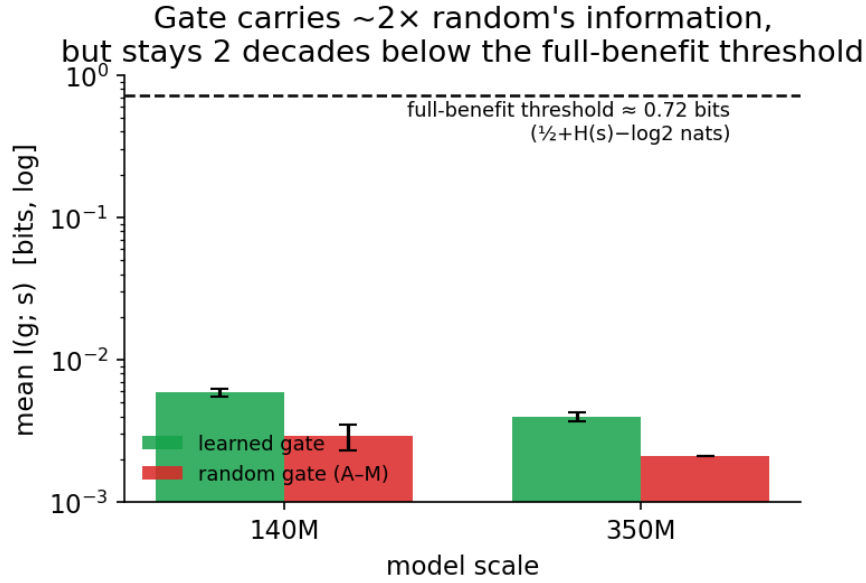


Fig 3: Gate mutual information vs the full-benefit threshold (log scale): two decades short at both scales.

No gate can reach the full-benefit threshold: the NSA-helps signal
is nearly information-free from token-local features

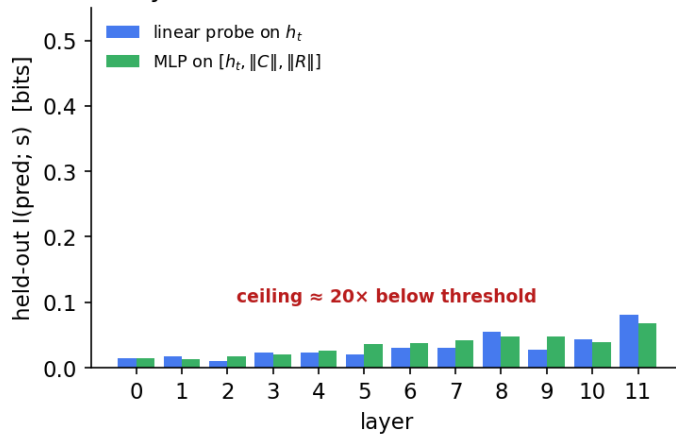


Fig 4: Optimal-probe ceiling per layer: ~ 0.034 bits, 21x below threshold.

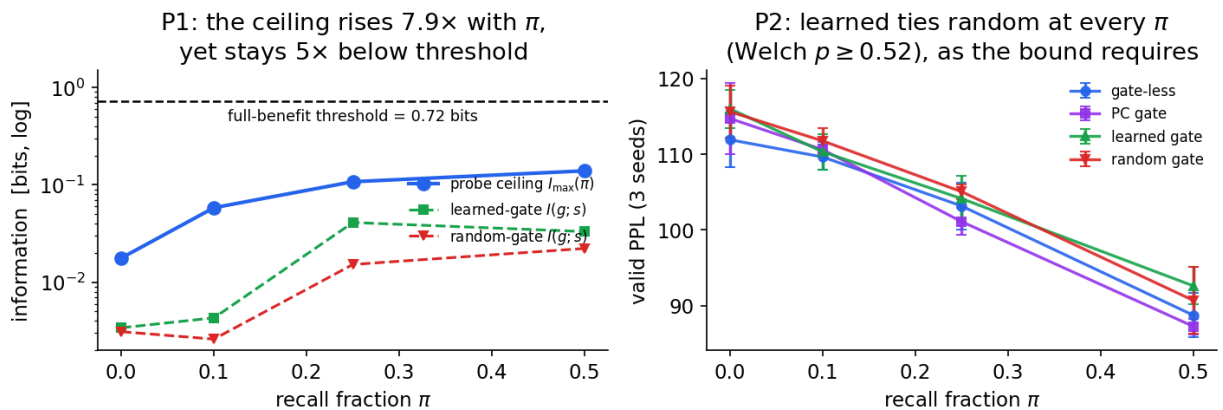


Fig 5: The executed routability phase diagram. Left: ceiling and gate MI vs recall fraction π (log scale) against the 0.72-bit threshold. Right: PPL by variant; learned ties random at every π .

Conclusion (abridged)

We set out to build a better router and ended up measuring why no such router can exist on this data. A gate needs about 0.72 bits about where attention helps; trained gates carry 0.005, and the oracle-label ceiling is 0.034, because attention helps everywhere and the exceptions are noise. The optimal policy is always-on, the gate-less baseline wins by construction, and the right response to routing absorption at this scale is no gate, plus data on which the question is worth asking. Probe the ceiling before training the router.

The constructive programme (pre-registered, now executed): mixture corpora with recall fraction π in $\{0, 0.1, 0.25, 0.5\}$ dial the routable structure up from zero. P1 confirmed: the probe ceiling $I_{\max}(\pi)$ rises $0.018 \rightarrow 0.058 \rightarrow 0.108 \rightarrow 0.139$ bits. P2 consistent: still 5x below threshold, and the learned gate ties random at every π , exactly as the bound requires. P3: no gate benefit anywhere; the recall task itself was not solved at this budget. P4 falsified: stream decorrelation does not raise the ceiling (0.094 vs 0.108 bits). Bonus: the parameter-free PC gate beats the learned gate pooled across the diagram and ties the gate-less baseline. Artifacts: `run_v6_queue.py`, `results/v6_routability_summary.md`.